

# Algebra I

## Equations, Functions & Graphing — Worked Examples

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Sample coursework material — for product demonstration purposes.

## Chapter 1 — Solving Linear Equations

A linear equation is solved by isolating the variable on one side using inverse operations, applied identically to both sides so the equation stays balanced.

**Worked example:** Solve  $3x + 7 = 22$  for  $x$ .

Step 1 — subtract 7 from both sides:  $3x = 15$ .

Step 2 — divide both sides by 3:  $x = 5$ .

Check:  $3(5) + 7 = 15 + 7 = 22$ . Correct.

The same approach extends to equations with variables on both sides, such as  $5x - 4 = 2x + 11$ : first collect the variable terms on one side (subtract  $2x$  from both sides to get  $3x - 4 = 11$ ), then isolate  $x$  as before ( $3x = 15$ , so  $x = 5$ ).

## Chapter 2 — Solving Quadratic Equations

A quadratic equation has the general form  $ax^2 + bx + c = 0$ . Many quadratics can be solved by factoring into two binomials whose product expands back to the original expression.

**Worked example:** Solve  $x^2 - 5x + 6 = 0$  by factoring.

We need two numbers that multiply to 6 and add to -5: those are -2 and -3.

Factored form:  $(x - 2)(x - 3) = 0$ , so  $x = 2$  or  $x = 3$ .

When a quadratic does not factor neatly, the quadratic formula always works:  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ . For  $x^2 - 5x + 6 = 0$ ,  $a = 1$ ,  $b = -5$ ,  $c = 6$ , giving the same two solutions,  $x = 2$  and  $x = 3$ .

## Chapter 3 — Systems of Linear Equations

A system of equations is a set of equations sharing the same variables, solved by finding values that satisfy all equations simultaneously. The substitution method solves one equation for one variable, then substitutes that expression into the other equation.

**Worked example:** Solve the system  $y = x + 2$  and  $2x + y = 11$ .

Substitute  $y = x + 2$  into the second equation:  $2x + (x + 2) = 11$ .

Simplify:  $3x + 2 = 11$ , so  $3x = 9$ , so  $x = 3$ .

Substitute back:  $y = 3 + 2 = 5$ . Solution:  $x = 3$ ,  $y = 5$ .

## Chapter 4 — Functions and Function Notation

A function assigns exactly one output to each input. Function notation  $f(x)$  reads as “ $f$  of  $x$ ” and represents the output of function  $f$  for input  $x$ .

**Worked example:** Given  $f(x) = 2x + 3$ , find  $f(4)$ .

Substitute  $x = 4$ :  $f(4) = 2(4) + 3 = 8 + 3 = 11$ .

Composite functions apply one function to the result of another. If  $f(x) = 2x + 3$  and  $g(x) = x^2$ , then  $f(g(2))$  first evaluates  $g(2) = 4$ , then  $f(4) = 11$ .

## Chapter 5 — Slope and Linear Graphs

The slope of a line measures its steepness: the ratio of vertical change to horizontal change between any two points,  $m = (y_2 - y_1) / (x_2 - x_1)$ .

**Worked example:** Find the slope between (1, 2) and (4, 11).

$$m = (11 - 2) / (4 - 1) = 9 / 3 = 3.$$

A line's equation can be written in slope-intercept form,  $y = mx + b$ , where  $m$  is the slope and  $b$  is the  $y$ -intercept — the point where the line crosses the  $y$ -axis.

## Chapter 6 — Exponents and Radicals

Exponent rules let us simplify expressions with repeated multiplication:  $a^m \times a^n = a^{m+n}$ , and  $(a^m)^n = a^{m \times n}$ .

**Worked example:** Simplify  $2^3 \times 2^2$ .

Add the exponents:  $2^{3+2} = 2^5 = 32$ .

**Worked example:** Evaluate  $\sqrt{144}$ .

$$12 \times 12 = 144, \text{ so } \sqrt{144} = 12.$$